A CHARACTERISATION OF TAUTNESS FOR MAPS OF WEAKLY ADMISSIBLE RINGS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Tautness of a continuous map of linearly topologized rings and the remarks following its definition. Same chapter, Definition definition-taut, invoked at proof lines 1328-1330. Gives one direction and openness of the closures used in the other.
- The set of topologically nilpotent elements is a closed ideal. Same chapter, Lemma lemma-topologically-nilpotent, invoked at proof lines 1332-1334. Shows that the closures consist of topologically nilpotent elements.

Source reference. Original lemma 0AMW: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1134–1134. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let $\varphi : A \to B$ be a continuous map of linearly topologized rings. We say φ is *taut*[1] if for every open ideal $I \subset A$ the closure of the ideal $\varphi(I)B$ is open and these closures form a fundamental system of open ideals.

Lemma 0.2. *Let $\varphi : A \to B$ be a continuous map of weakly admissible topological rings. The following are equivalent*

(1) *φ is taut,*
(2) *for every weak ideal of definition $I \subset A$ the closure of $\varphi(I)B$ is a weak ideal of definition of B and these form a fundamental system of weak ideals of definition of B.*

Proof. The remarks following Definition 0.1 show that (2) implies (1). Conversely, assume φ is taut. If $I \subset A$ is a weak ideal of definition, then the closure of $\varphi(I)B$ is open by definition of tautness and consists of topologically nilpotent elements

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[1]This is nonstandard notation. The definition generalizes to modules, by saying a linearly topologized A-module M is A-taut if for every open ideal $I \subset A$ the closure of IM in M is open and these closures form a fundamental system of neighbourhoods of 0 in M.

by Lemma 0AMW. Hence the closure of $\varphi(I)B$ is a weak ideal of definition. Furthermore, by definition of tautness these ideals form a fundamental system of open ideals and we see that (2) is true. $\square$